

Path Planning with Cable-obstacles Avoidance for a Tethered Mobile Robot in Unstructured Environments

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Tethered mobile robots for lunar and planetary exploration can extend the scientific return from unexplored areas that conventional mobile robots cannot traverse, such as steep slopes near lunar craters, lava tubes, or recurring slope lineae (RSL) on Mars. These robots use a cable for stable communication and power supply, improving maneuverability in unstable and challenging terrains. However, the cable can limit the robot's range of motion and cause entanglement or contact with obstacles. These make the path planning difficult. This study presents a path-planning method that focuses on cable-obstacles avoidance for safer and longer exploration. The method modifies the pre-planned path based on its curvature, distance from the cable anchoring point, and distance from obstacles. The proposed method was tested using an in-house simulator for a tethered robot with a geometry-based cable model. The results demonstrated that our method improved cable-to-obstacles contact avoidance by about 74 % compared to a heuristic search algorithm in two-dimensional environments.

Key Words: Tethered Mobile Robot, Path Planning, Lunar Exploration, Artificial Potential Fields

1. Introduction

Tethered mobile robots have recently emerged as a promising solution for lunar and planetary exploration, enabling access to challenging areas that conventional mobile robots cannot traverse, such as steep slopes near lunar craters, lava tubes, or recurring slope lineae (RSL) on Mars. These robots utilize a cable for stable communication and power supply, which significantly improves their maneuverability in unstable and challenging terrains. Moreover, the tethered design ensures a reliable connection to a base station, enabling long-duration missions with continuous data transmission. The ability to navigate and operate the tethered robot in such challenging terrains increases the scientific return from these missions, allowing researchers to collect valuable data and conduct experiments in previously inaccessible areas. This enhanced navigational capability, in turn, helps advance our understanding of the geological and environmental processes on these celestial bodies.

However, the use of a cable also introduces certain challenges for path planning and navigation. The cable can limit the robot's range of motion, causing entanglement or contact with obstacles, making the path-planning process more complex and demanding.

To overcome these challenges, Igarashi et al. have revealed that path planners for a tethered mobile robot must be aware of differences in cable placement, even though the robot's position is the same.¹⁾ They have proposed a solution to build a manifold from a grid map. The manifold is a graph that can have an overlap, i.e., two graph nodes can be assigned to the same grid. Constructing a manifold can distinguish the cable placements based on the Manhattan distance from the cable anchoring point. Kim et al. have pointed out that the cable placement cannot be identified by considering the simple metric (Manhattan distance).²⁾ They have proposed building a homotopy-augmented graph. The graph encodes the cable arrangement into a string that represents the cable's routing orientation for

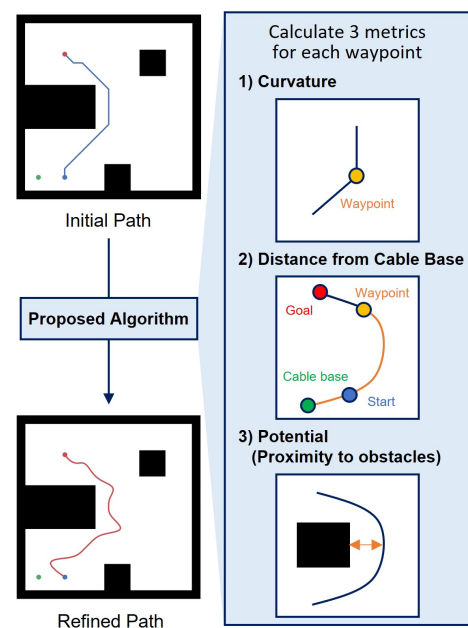


Fig. 1. Concept of the proposed path planning algorithm for cable-obstacles avoidance. To achieve the avoidance, our algorithm enhances the cornering behavior around obstacles by refining the pre-planned path based on the three metrics of each waypoint: 1) curvature, 2) distance from cable base, and 3) potential as proximity to obstacles. The red, green, and blue dots denote the goal point, cable base, and start point. The orange dot represents a waypoint of interest.

each obstacle, as clockwise or counterclockwise. This approach performs well but does not consider the cable-obstacles contact or snagging.

In this study, we present a path-planning method for tethered mobile robots, focusing on cable-obstacles avoidance. Our method emphasizes the cornering behavior of the robot, which is a critical aspect when navigating around obstacles (Figure 1). During sharp turns around obstacles, the cable often comes

close to the obstacle, increasing the risk of cable-to-obstacles contact. This contact can lead to entanglement and potential damage to the cable, ultimately jeopardizing the success of the mission. To reduce this risk, the proposed method calculates the following three metrics: the curvature, the distance from the cable base, and the potential of waypoints on a pre-planned path. The method then modifies the path to allow the robot to make larger turns around high-curvature points. By facilitating smoother turns, the robot can maintain a safer distance between the cable and obstacles, reducing the likelihood of contact. Consequently, this path-planning method enables tethered mobile robots to carry out safer and longer exploration missions by minimizing the risk of cable entanglement and damage.

In order to evaluate the effectiveness of the proposed algorithm, we conducted a series of experiments using an in-house simulator for a kinematic robot model, which includes a geometry-based cable model. Then, the proposed method was compared to a heuristic search algorithm in various environments using the in-house simulation experiment. The results of these experiments demonstrated that the proposed method significantly reduced cable-to-obstacles contact; in particular, the proposed method achieved a 74% improvement in contact avoidance over the heuristic search algorithm. These findings suggest that our path-planning method could enhance the safety and longevity of tethered mobile robots during lunar and planetary exploration missions.

The remainder of this paper is organized as follows. Section 2 presents a path-planning method with cable-obstacles avoidance for a tethered mobile robot. Section 3 introduces the robot and cable model used in our in-house simulator. Section 4 shows the simulation results and discusses the performance of the proposed method. Finally, Section 5 concludes this paper and summarizes future research directions.

2. Path Planning Algorithm with Cable-obstacles Avoidance

This section presents a path-planning method for a tethered mobile robot with cable-obstacles avoidance. We first outline the proposed method, and the following several subsections explain the detailed components: construction of artificial potential fields, calculation of curvature for discrete points, and obstacle-aware waypoint modification method.

2.1. Algorithm Overview

Our algorithm adjusts a pre-planned path in areas where the curvature is high or where the waypoint is far from the cable anchoring point. It displaces the waypoints based on the potential, which can be defined in an artificial potential field. Algorithm 1 describes our method in detail.

The following subsections describes the components of the algorithm.

2.2. Algorithm: Step 1 — Generating the initial path

Given that the environment is a grid map of size $H \times W$, the robot's workspace is effective $(H - 2) \times (W - 2)$ because of its surrounding obstacles. In this environment, we use the A^* algorithm³⁾ to obtain the initial path connecting the predetermined start point and the randomly generated goal point.

Algorithm 1: Path planning with cable-obstacles avoidance

Input: Pre-planned path τ , Attractive potential U_{att} , Repulsive potential U_{rep} , Mixing factor α , Cable base (x_b, y_b) , Tuning parameter η

Output: Refined path τ_{new}

```

1  $\tau_{new}[0] \leftarrow \tau[0]$ 
2  $\hat{k} \leftarrow \text{CALCNORMCURVATURE}(\tau)$ 
3  $\hat{d} \leftarrow \text{CALCNORMDISTANCEFROMCABLEBASE}(\tau, x_b, y_b)$ 
4  $\hat{U} \leftarrow \text{CALCNORMPOTENTIAL}(U_{att}, U_{att})$ 
5 for  $i \leftarrow 1$  to  $\text{len}(\tau) - 1$  do
6    $P \leftarrow \tau[i]$ 
7    $n \leftarrow \text{FINDNORMALUNITVECTOR}(\tau[i-1], \tau[i], \tau[i+1])$ 
8   if  $\hat{k}[i-1] \neq 0$  then
9      $P_{new} \leftarrow P + \eta \cdot \hat{d}[i] \cdot \hat{k}[i-1] \cdot \hat{U} \cdot \frac{n}{\|n\|}$ 
10  else
11     $P_{new} \leftarrow P$ 
12  end
13   $\tau_{new}[i] \leftarrow P_{new}$ 
14 end
15  $\tau_{new}[\text{len}(\tau) - 1] \leftarrow \tau[\text{len}(\tau) - 1]$ 
16 return  $\tau_{new}$ 

```

2.3. Algorithm: Step 2 — Calculating the three metrics for all waypoints

Our algorithm calculates the three metrics for each waypoint: curvature, distance from the cable base, and potential (used as a proximity to the obstacles).

Curvature: A curvature can generally be defined for a curve that is twice differentiable and continuous. However, for paths consisting of a finite number of waypoints, curvature must be defined in a different way. We define curvature as the reciprocal of the circle's radius passing through the waypoint and two adjacent points on the path. Let $A(x_0, y_0)$, $B(x_1, y_1)$, and $C(x_2, y_2)$ be three consecutive points. To calculate the curvature, we first define the determinant D as

$$D := \det \begin{pmatrix} x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \\ 1 & 1 & 1 \end{pmatrix}. \quad (1)$$

When $D = 0$ holds, the three points are on the same line, and we suppose the curvature, in this case, is 0. When $D \neq 0$ holds, the radius r of a circle passing through the points A, B, and C can be expressed as

$$r = \sqrt{(x_0 - h)^2 + (y_0 - k)^2}, \quad (2)$$

where (h, k) denotes the coordinates of the center of the circle and is written as

$$\begin{cases} h = \frac{1}{D} \sum_{i=0}^2 (x_i^2 + y_i^2)(y_{(i+1) \bmod 3} - y_{(i-1) \bmod 3}) \\ k = \frac{1}{D} \sum_{i=0}^2 (x_i^2 + y_i^2)(x_{(i-1) \bmod 3} - x_{(i+1) \bmod 3}) \end{cases}. \quad (3)$$

Thus, the curvature around point B is expressed as

$$\kappa_B = \text{sign}((x_1 - x_0)(y_2 - y_0) - (y_1 - y_0)(x_2 - x_0)) \cdot \frac{1}{r}, \quad (4)$$

where we define the curvature to be negative if the cross product of the two vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is negative. The curvature of each waypoint on the initial path will be normalized as

$$\hat{\kappa}_i = \left| 2 \cdot \frac{\kappa_i - \kappa_{\min}}{\kappa_{\max} - \kappa_{\min}} - 1 \right|, \quad i \in [0, n-1]. \quad (5)$$

Distance from Cable Base: The tethered mobile robot can perform a cable extraction movement close to the cable base. Conversely, the robot's motions to avoid contact become more limited as it moves away from the base. Here we define the distance from the cable base and describe how we normalize it. The total path length L is written as

$$L = \sum_{i=0}^{n-2} \|\mathbf{x}_i - \mathbf{x}_{i+1}\|, \quad (6)$$

where $\mathbf{x}_i$ is the waypoint on the path. And the cumulative distance $D_{\text{cum}, i}$ is expressed as

$$D_{\text{cum}, i} = \sum_{j=0}^i \|\mathbf{x}_j - \mathbf{x}_{j+1}\|, \quad i \in [0, n-2]. \quad (7)$$

Using the Euclidean distance between cable base $\mathbf{x}_b$ and start point $\mathbf{x}_0$, the distance from cable base can be expressed as $\|\mathbf{x}_0 - \mathbf{x}_b\| + D_{\text{cum}, i}$ and we define its normalized distance $\hat{d}_i$ as

$$\hat{d}_i = 1 - \frac{\|\mathbf{x}_0 - \mathbf{x}_b\| + D_{\text{cum}, i}}{L}. \quad (8)$$

The normalized distance ranges from $[0, 1]$ and takes on larger value as it gets closer to the cable base.

Artificial Potential Fields: The artificial potential field (APF) method constructs the environment with two forces: a repulsive force that keeps the robot away from obstacles and an attractive force that drives the robot toward the target point. We used this APF method in order to determine how distant from the obstacles a waypoint on a pre-planned path is. The attractive force can be expressed as:

$$U_{\text{att}}(x, y) = \frac{1}{2} K_{\text{att}} \sqrt{(x - x_{\text{goal}})^2 + (y - y_{\text{goal}})^2}, \quad (9)$$

where K_{att} is a positive gain on the attractive potential. The repulsive potential, on the other hand, can be written as:

$$U_{\text{rep}}(x, y) = \begin{cases} K_{\text{rep}} \cdot \exp\left(-\frac{(x-o_x)^2 + (y-o_y)^2}{2\sigma^2}\right) & \text{if } \rho(x, y) \leq \rho_0, \\ 0 & \text{otherwise,} \end{cases} \quad (10)$$

where K_{rep} is a positive gain on the repulsive potential and $\rho(x, y)$ is the distance to the nearest obstacle. If the distance is lower than or equal to ρ_0 , the repulsive force is generated.

For all grids of an environment, the potential is computed as the sum of the two above potentials:

$$U(x, y) = U_{\text{att}}(x, y) + U_{\text{rep}}(x, y). \quad (11)$$

The potential can be normalized using

$$\hat{U}_i = \frac{U_i - U_{\min}}{U_{\max} - U_{\min}}, \quad i \in [0, n-1]. \quad (12)$$

Figure 2 shows an image of the artificial potential field using the above equations.

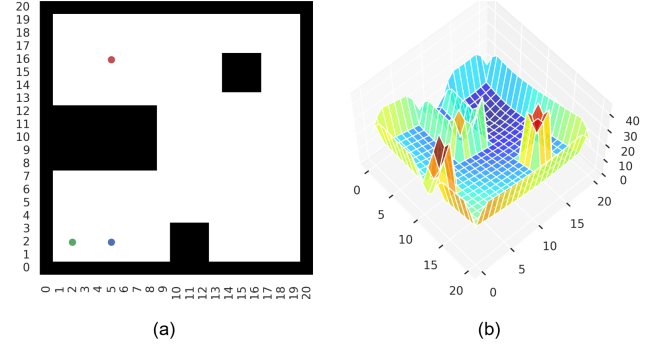


Fig. 2. Typical artificial potential field. (a) The dot markers denote the start point (blue), goal point (red), and cable base (green). (b) Potential field corresponding to environment A. The area surrounding the obstacle is elevated and inclined toward the goal.

Algorithm 2: Find Optimal Tuning Parameter

Input: Pre-planned path τ , Attractive potential U_{att} , Repulsive potential U_{rep} , Cable base (x_b, y_b) , Obstacles $\mathcal{O}$, Tuning parameter $\eta_{\min}, \eta_{\max}, \eta_{\text{step}}$

Output: Optimal tuning parameter η^*

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1  $C_{\min} \leftarrow \infty$ 
2  $\eta^* \leftarrow 0$ 
3 for  $\eta \leftarrow (\eta_{\min}, \eta_{\max}, \eta_{\text{step}})$  do
4    $\tau_{\text{mod}} \leftarrow \text{MODIFYPATH}(\tau, U_{\text{att}}, U_{\text{rep}}, x_b, y_b, \eta)$ 
5    $\tau_{\text{smoothed}} \leftarrow \text{SMOOTHPATH}(\tau_{\text{mod}})$ 
6   if not  $\text{ISPATHINTERSECTOBSTACLES}(\tau_{\text{smoothed}}, \mathcal{O})$  then
7      $C_{\text{tot}} \leftarrow \text{CALCCLEARANCE}(\tau_{\text{smoothed}}, \mathcal{O})$ 
8     if  $C_{\text{tot}} < C_{\min}$  then
9        $C_{\min} \leftarrow C_{\text{tot}}$ 
10       $\eta^* \leftarrow \eta$ 
11    end
12  end
13 end
14 return  $\eta^*$ 

```

2.4. Algorithm: Step 3 — Finding an optimal path

Our method displaces the position of each waypoint with three previously described metrics: curvature, distance from the cable base, and potential. In this subsection, we introduce a tuning parameter η that determines the inflation distance for waypoints to be displaced (Algorithm 1). Denoting the candidate path with the tuning parameter η in τ_η , the value of η takes a non-negative real number, and we optimize it by considering the following problem:

$$\underset{\eta}{\text{maximize}} \quad C(\tau_\eta) \quad (13a)$$

$$\text{subject to} \quad \tau_\eta \notin \mathcal{O} \quad (13b)$$

We want to find an optimal η that satisfies the minimum potential among the candidates of refined paths and holds the non-intersection condition with the obstacles in the environment. To solve this problem, we use the linear search as shown in Algorithm 2. After finding an optimal η , our algorithm smoothes the path based on the modified waypoints using B-spline curve.⁴⁾ The differences in the smoothed modified paths for different η

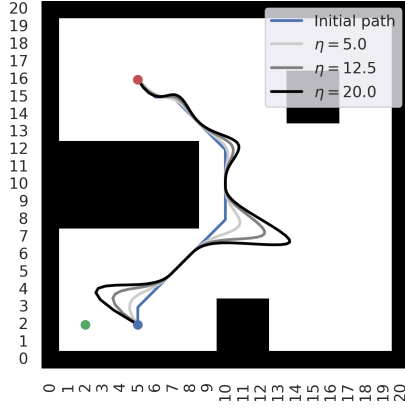


Fig. 3. Difference in the modified paths for different tuning parameter η .

are shown in Figure 3.

3. Simulation Model

As introduced in Section 1, the proposed method is verified using an in-house simulator for the tethered robot. The simulation model consists of a kinematic-based robot model and a geometry-based model for the cable.

3.1. Kinematic Robot Model

This subsection presents a robot model with a kinematics-based robot mode. The robot follows a path generated by our method using the pure pursuit algorithm,⁵⁾ and counts the cable-obstacle and cable-robot contacts on its way to a goal.

We used the velocity motion model,⁶⁾ as shown in Figure 4 for ease of implementation. We can directly input the command of the velocity and angular velocity to the model. The robot state can be written as

$$\mathbf{x} = [x, y, \theta]^T. \quad (14)$$

The control input is set to

$$\mathbf{u}_t = [v_t, \omega_t]^T, \quad (15)$$

where v_t is the translational velocity and ω_t is the angular velocity. By differentiating Equation 14 with respect to time, we obtain the following expression for the velocity:

$$\dot{\mathbf{x}} = [v \cos \theta, v \sin \theta, \omega]^T. \quad (16)$$

The relative posture of the robot between time $t - 1$ and next

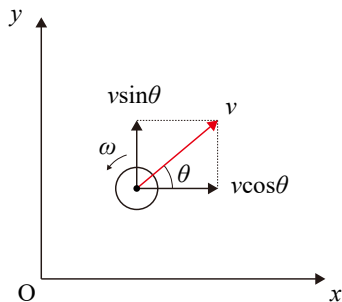


Fig. 4. Kinematic robot model. The translational velocity v and the rotational velocity ω are the control command into the model.

time t is as follows:

$$\begin{aligned} \theta_t &= \theta_{t-1} + \int_0^{\Delta t} \omega_t dt \\ &= \theta_{t-1} + \omega_t \Delta t, \end{aligned} \quad (17)$$

where Δt is the time step. From Equation 16, we obtain

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} \int_0^{\Delta t} v_t \cos(\theta_{t-1} + \omega_t t) dt \\ \int_0^{\Delta t} v_t \sin(\theta_{t-1} + \omega_t t) dt \end{bmatrix}. \quad (18)$$

When $\omega_t = 0$ holds, we find

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} \Delta t v_t \cos \theta_{t-1} \\ \Delta t v_t \sin \theta_{t-1} \end{bmatrix}, \quad (19)$$

$\omega_t \neq 0$, we get

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \frac{v_t}{\omega_t} \begin{bmatrix} \sin(\theta_{t-1} + \omega_t \Delta t) - \sin \theta_{t-1} \\ -\cos(\theta_{t-1} + \omega_t \Delta t) + \cos \theta_{t-1} \end{bmatrix}. \quad (20)$$

A robot with this model follows a generated path with the translational velocity calculated by feedback control and the rotational velocity computed by the pure pursuit algorithm.

3.2. Geometry-based Cable Model

A cable is modeled as a chain of nodes; when a cable node is pulled, the rest of the nodes follow it. This method is called the geometry-based or position-based model, often used in computer graphics. The node positions are updated as follows.

1. Apply the robot motion to the end node.
2. Check if the robot is within the total length of the cable.
3. Update the remaining nodes to follow the end node displaced in Step 1.
4. Detect collisions and correct the node positions if necessary.

To update the positions of follower nodes in Step 3, we used the Follow-the-leader algorithm.^{7,8)} Given that one end of the cable termed as $\mathbf{x}_{n-1}$ is attached to the robot where n is the number of cable nodes and the node $\mathbf{x}_{n-1}$ moves from $\mathbf{x}_{n-1, \text{old}}$ to $\mathbf{x}_{n-1, \text{new}}$ along with the movement of the robot, the positions of the remaining nodes can be calculated as

$$\begin{aligned} \mathbf{x}_{i, \text{new}} &= \mathbf{x}_{i+1, \text{new}} + \frac{(\mathbf{x}_{i, \text{old}} - \mathbf{x}_{i+1, \text{new}})}{\|\mathbf{x}_{i, \text{old}} - \mathbf{x}_{i+1, \text{new}}\|} \\ &\quad (i = n-2, n-3, \dots, 0). \end{aligned} \quad (21)$$

In the above equation, the cable is assumed to be not anchored to the environment.

Figure 5 illustrates the case where the cable is fixed. Note that the cable is placed on the ground, and no gravity acts on it. Computing complete displacement in this assumption requires the following four steps: (a) the robot moves in the direction of the red arrow, and the end nodes follow; (b) the rest of the nodes, except the fixed end, follow using Equation 21; (c) consider the fixed end to have moved a zero distance, and update the nodes except the free end using Equation 21; (d) the final displacements are computed as the midpoints between (b) and (c) for all nodes except two ends.

Cable-robot Entanglements and Detection: In our model, the cable is assumed to be composed of n number of sequentially-collocated nodes $\{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{n-1}\}$, where $\mathbf{x}_0$ is the cable base and $\mathbf{x}_{n-1}$ is attached to the robot (Figure 6). Here, let r_{rbt} be the

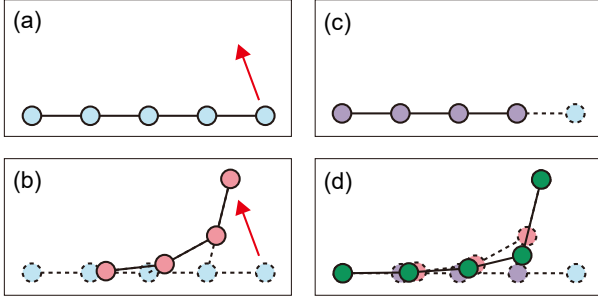


Fig. 5. Procedure for updating cable nodes when one end is fixed. (a) The left end is anchored, and the right end is displaced by a robot in the direction of the red arrow. (b) All nodes except the anchored node follow the movement of the robot node. (c) Now consider the movement of nodes without moving the robot while the anchored node is displaced. The positions of all nodes are not updated. (d) After applying the red-arrow displacement, the final cable position can be found by computing the midpoints of (b) and (c).

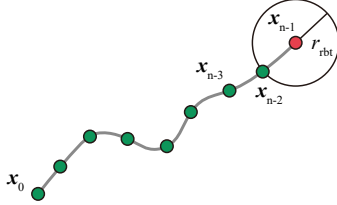


Fig. 6. Cable entanglement around a circular robot. The green and red dots represent the n cable nodes. The black circle with the line segment indicates the robot's position and heading. The node x_0 is anchored to the ground, whereas the node x_{n-1} is attached to the center of the circular robot. Our algorithm detects the cable-robot entanglement based on the Euclidean distance between the red node and $n-1$ green nodes.

radius of the circular robot and l_{seg} be the length of one cable segment, i.e., the total cable length equals $(n-1)l_{\text{seg}}$.

Assuming $r_{\text{rft}} = l_{\text{seg}}$, the cable-robot collision occurs in the case if the following condition holds:

$$\|x_{n-1} - x_i\| \leq r_{\text{rft}} \quad (i = 0, 1, \dots, n-3). \quad (22)$$

Cable-obstacle Contact and Detection: We used a well-known collision detection method in computer graphics called axis-aligned bounding box (AABB).⁹⁾ This method benefits from its computational simplicity $O(1)$. The intersection of two AABBs shown in Figure 7 can be determined by

$$(x_{\text{Amin}} \leq x_{\text{Bmax}} \wedge x_{\text{Amax}} \geq x_{\text{Bmin}}) \\ \wedge (y_{\text{Amin}} \leq y_{\text{Bmax}} \wedge y_{\text{Amax}} \geq y_{\text{Bmin}}).$$

Thereby, for n cable nodes and M obstacles in the environment, all these intersections can be computed in at most $O((n-1)M)$.

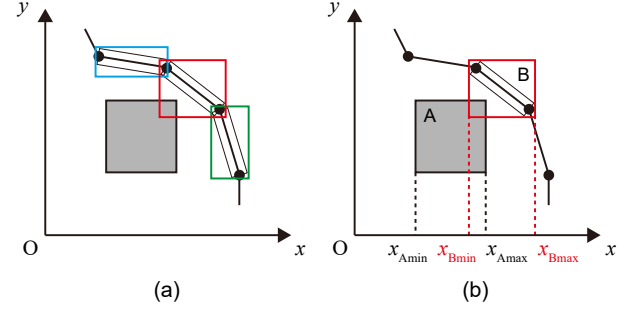


Fig. 7. Overlap determination of two AABBs. (a) The gray square denotes an AABB for an obstacle. The black dots are cable nodes, and the black lines connecting the dots represent cable segments. Black rectangles around cable segments indicate a specific cable radius. Blue, red, and green rectangles also represent AABBs for each cable segment. (b) We can detect the overlap of two AABBs, by comparing the coordinates of the four edges of two AABBs.

4. Simulation Results and Discussion

This section demonstrates the proposed method through simulation in environments with different obstacle densities. The cable is assumed to be pulled out from the unreeling mechanism as the robot moves. We evaluated the performance based on two metrics: path length and number of times of cable-obstacle contact per minute. All simulations were executed on an Intel(R) Core(TM) i7-12700 CPU clocked at 2.10 GHz and Python 3.8.

4.1. Implementation Details

Table 1 presents the simulation parameters and their values. Our simulations were performed on a 20×20 grid map. The initial position of the tethered robot was set to (5, 2) and the cable base to (2, 2) in Cartesian coordinates as described in Figure 8. Obstacles were randomly generated using the maximum number of obstacles and maximum obstacle size as inputs. The goal point was randomly set to be at least 10 Manhattan distances away from the start point.

Under these settings, the initial path was generated using the A* algorithm.³⁾ The proposed path modification method was applied to the generated paths. Then, the robot with the kinematic model described in Section 3 tracked both the initial and the refined path (Figure 8). We counted the number of contacts until the robot reached the goal.

4.2. Evaluation Metrics

We used the following four evaluation metrics: the first two are used for comparison with the naive approach, and the latter two are used for comparison with the Voronoi Diagram.

Table 1. Simulation parameters and values.

Parameter	Value	Description
K_{att}	2.0	Gain of attractive potential
K_{rep}	5.0	Gain of repulsive potential
ρ_0	2.0	Threshold distance for repulsive force
σ	3.0	Variance in Gaussian function
$d_{\text{MAN,min}}$	10.0	Min. Manhattan distance (start and goal)
H	20	Height (longitudinal size) of the map
W	20	Width (lateral size) of the map
n	30	Number of cable nodes

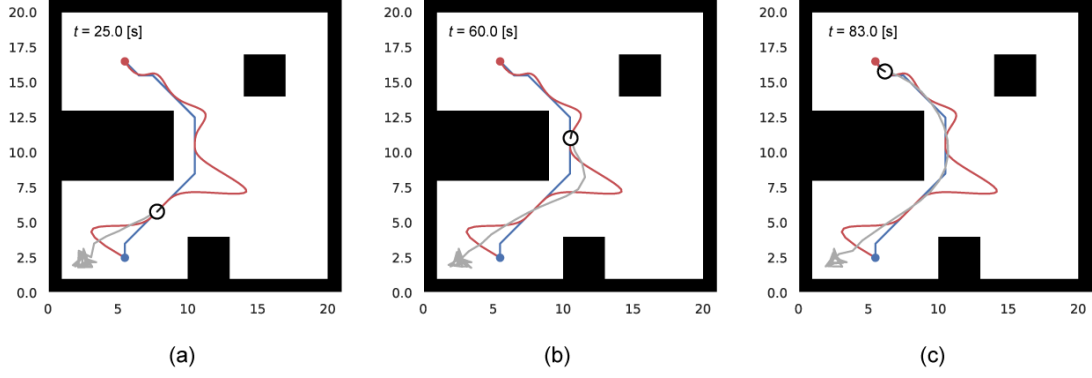


Fig. 8. Typical path tracking procedure: initial path (red), cable state (gray), and robot trajectory on the refined path (purple). The blue-circle marker denotes the start position, whereas the red circle denotes the goal. The circle with the line segment indicates the robot's position and heading.

Path Length: The refined path becomes longer than the pre-planned path because our method inflates the input path. We quantitatively evaluate the trade-off between path length and contact avoidance performance. The length is calculated by the sum of the Euclidean distance of two adjacent waypoints as

$$L(\tau) = \sum_{i=0}^{m-2} \|p_i - p_{i+1}\|, \quad (23)$$

where m is the number of points composing the path τ .

Number of Cable-obstacles Contact: We count the number of times cable-obstacles contact is detected from the start to the goal of the robot. The simulation runs in 0.1-second time steps. To treat the different simulations equally, the contact time is calculated by setting the time from start to goal as 60 seconds.

Fréchet Distance: Fréchet distance¹⁰⁾ is used to evaluate the similarity between two paths. It is defined as the shortest pair distance between both paths that can completely traverse two paths. Mathematically, using the two paths τ_1 and τ_2 and the continuous maps $f : [0, 1] \rightarrow \tau_1$ and $g : [0, 1] \rightarrow \tau_2$, it is expressed as follows:

$$FD(\tau_1, \tau_2) = \min_{f,g} \max_{t \in [0,1]} \|f(t) - g(t)\|. \quad (24)$$

We used the discredited version of this metric to evaluate the similarity between the generated path and the path in Voronoi Diagram.

Average Angle Difference: Average angle difference is a measure of the directional similarity of two paths. This measure is calculated by comparing the directional difference of each node of two paths.

Calculate the angle for discrete points on the two paths:

$$\begin{cases} \theta_{1,i} = \arctan 2((\tau_1)_{i+1,y} - (\tau_1)_{i,y}, (\tau_1)_{i+1,x} - (\tau_1)_{i,x}) \\ \theta_{2,i} = \arctan 2((\tau_2)_{i+1,y} - (\tau_2)_{i,y}, (\tau_2)_{i+1,x} - (\tau_2)_{i,x}) \end{cases} \quad (25)$$

Then calculate the angle difference:

$$\Delta\theta_i = \min\{|\theta_{1,i} - \theta_{2,i}|, 2\pi - |\theta_{1,i} - \theta_{2,i}|\} \quad (26)$$

Table 2. Performance comparison between the naive approach and the proposed method.

	Naive	Ours
Path Length [-]	13.67 ± 3.21	16.59 ± 4.51
Cable-Obstacle Contact [times/min]	9.30 ± 10.27	2.40 ± 5.21
Imp. of Avoidance [%]	-	74.16 ± 26.45

Finally, we obtain

$$AAD(\tau_1, \tau_2) = \frac{1}{m-1} \sum_{i=0}^{m-2} \Delta\theta_i, \quad (27)$$

We evaluate how precisely one path follows the other by computing the directional difference of each node and finding its average. This measure helps evaluate how similar the shapes and orientations of the curves are, but it does not consider position or scale. Therefore, it is combined with other measures (e.g., in this study, the Fréchet distance) for a more comprehensive similarity evaluation.

4.3. Quantitative Evaluation

We simulated 20 random environments and compared the performance of the naive method. Two evaluation metrics were used: cable length and cable obstacle contact.

Results in random environments: The results in Table 2 demonstrate that our method enhances contact avoidance performance while increasing cable length. Consequently, despite a slight increase in path lengths, the proposed method effectively avoids cable-obstacle contact. Furthermore, it offers more consistent outcomes regarding cable-obstacle contact across different obstacle densities. This analysis can evaluate the effectiveness and stability of the proposed method.

Similarity with Voronoi Diagram: Our method is similar to the Voronoi diagram in finding the path with the minimum potential to avoid contact with obstacles. A Voronoi diagram divides space into regions by connecting the points farthest from obstacles. In the robot path planning problem, safe paths are generated by assigning paths on these boundaries.¹¹⁾ In this subsection, we discuss their similarity using the Fréchet distance and the average angle difference.

Table 3 shows the similarities between three types of paths. We can see the highest similarity between Ours and Naive. This

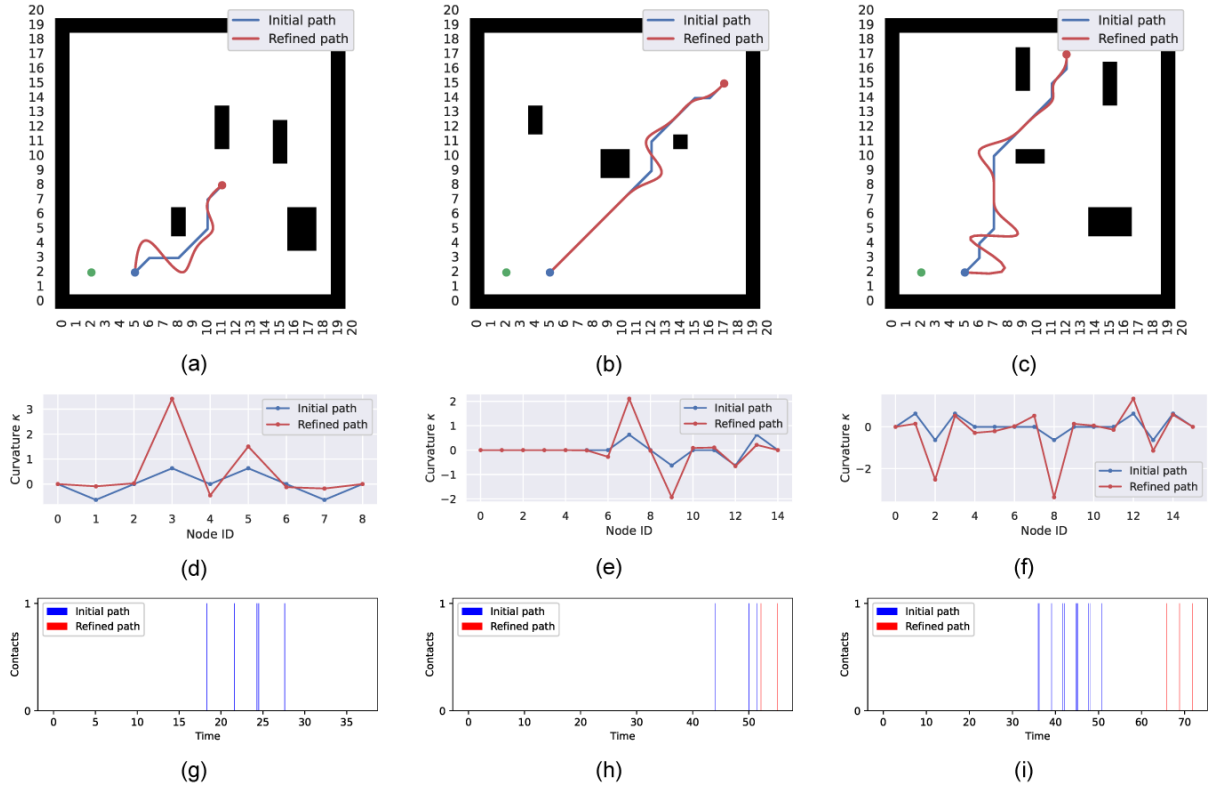


Fig. 9. Typical results of our path planning algorithm with cable-obstacles avoidance in environments with different obstacle densities: (a-c) Comparison of initial and refined paths. The three circle markers denote the start position (blue), goal position (red), and cable base (green). The two curves represent the initial path (blue) and the refined path (red). (d-f) Change in curvature along the arc length of the two paths. The x-coordinate 0 corresponds to the start point. Our method inflates the high curvature points of the initial path. (g-i) Time at which cable-obstacle collision is detected.

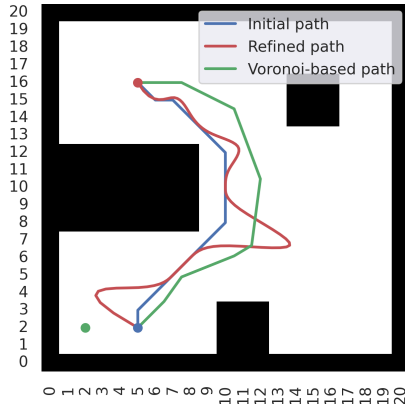


Fig. 10. Visual comparison of three methods: naive approach, proposed method, and Voronoi diagram-based method.

Table 3. Comparison of similarity for three path planning methods.

Pair	Frechet Distance	Average Angle Difference
Ours — Naive	1.26 ± 0.56	0.29 ± 0.14
Naive — Voronoi	3.04 ± 1.58	0.59 ± 0.16
Voronoi — Ours	2.57 ± 1.59	0.41 ± 0.18

is natural since the proposed method is derived from a naive approach. On the other hand, the similarity between Naive and Voronoi is lower than that between Ours and Naive. This indicates that the naive and Voronoi diagram-based approaches produce different types of paths. The similarity between Voronoi and Ours shows that the proposed method generates paths that are somewhat different from the Voronoi diagram-based method

but still more similar to the naive approach. Therefore, the proposed method generates more aggressive paths than the Voronoi diagram-based method in areas where the effect of cable-obstacles contact is less significant.

4.4. Qualitative Evaluation

Figure 9 shows three typical results. The top part of the figure (a-c) shows the paths before and after the modification; the middle part (d-f) shows the change in path curvature; and the bottom part (g-i) shows the time the contact occurred. Below, we qualitatively evaluate the results in terms of the change in path shapes and contact characteristics.

Change in Path Shape: The proposed method modifies the initial path by considering three factors: the curvature of the path, the distance from the cable base, and the potential. We will discuss each of these factors in turn. First, as seen in Figure 9 (d-f), the high curvature of the initial path is inflated, indicating that the method adjusts the path based on its curvature. Second, Figure 9 (a-c) demonstrates that the amount of modification increases as the path gets closer to the cable base, highlighting the influence of the distance from the cable base on the modification process. Lastly, the potential-based modifications can be observed by comparing the initial and modified paths. The modified path is further away from the obstacles than the initial path, indicating that the potential contributes to shaping the final path.

Characteristics of Cable-obstacles contact: The occurrence and nature of cable-obstacle contact in different scenarios can

be analyzed by examining Figure 9 (g-i). In Figure 9 (g), we observed total contact avoidance; any contact present in the initial path is fully eliminated by the modification. However, it is important to note that such outcomes are achievable only within a restricted set of environments. In Figure 9 (h), there is a slight decrease in the number of contacts, and the contact timing is delayed in the modified path compared to the initial path. This suggests that once the robot pulls out the cable to a certain extent, some portions of the robot's movement become uncontrollable, inevitably leading to contact. Lastly, Figure 9 (i) presents a case where the number of contacts is significantly reduced, with contact occurring toward the end of the simulation. This suggests that there are segments of the cable that remain uncontrollable in the sense that a tethered mobile robot cannot directly influence them through its movement. These observations demonstrate the effectiveness of the proposed method in reducing cable-obstacle contact while highlighting the challenges and limitations in achieving perfect contact avoidance in certain situations.

5. Conclusion

This paper presented a path-planning method specifically designed for tethered mobile robots. The method takes into consideration the avoidance of cable-obstacles while providing safer and longer exploration capabilities. By modifying the pre-planned path based on its curvature, distance from the cable base, and potential, our proposed method effectively addresses the challenges posed by the presence of a cable. We tested the method using the in-house simulator for the kinematic robot model with the geometry-based cable model. Our results demonstrated that the proposed method significantly improves cable-to-obstacles avoidance by about 74 % compared to a heuristic search algorithm. However, complete avoidance of the contact remains an open issue. This issue could be solved by analyzing which parts of the cable become less maneuverable as the distance traveled by the robot increases.

Furthermore, we have shown quantitatively that the paths generated by our method are similar to the paths on the Voronoi diagram, using the Frechet distance and the average angle difference. This can be interpreted to mean that our method gen-

erates more aggressive paths than the Voronoi diagram.

Our method has only been validated in a 2D environment, which represents a significant gap between the current capabilities of our method and its potential use in future extraterrestrial exploration missions. To bridge this gap, our future work will focus on:

- Extending our method to a 2.5D environment that accounts for inclination,
- Incorporating a dynamic cable model, and
- Conducting experimental validation on a physical robot.

By addressing these future research directions, we aim to enhance the effectiveness of tethered robot path planning, ultimately facilitating more productive exploration missions.

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